Manish Kumar Goyal, Chair Professor,
& Former Dean (2020-2025)
JSPS Fellow, Commonwealth Fellow, Erasmus Fellow,
P.E.(Int), Indo-US WARI Fellow, Inspire Fellow
Indian Institute of Technology, Indore-
Khandwa Road, Simrol-453552, M.P. India
Email: mkgoyal@iiti.ac.in, Phone: +91 9435880989

1. Research Interests: Water, Climate change, Modelling, Hazard, Remote Sensing, Data mining

2. Research Publication Summary

Books	12
Book Chapters	35
Peer Reviewed Journals	155
Conferences proceeding	60
Patent	17
Total	**274**

Bibliometric Data

Total Citations (Google Scholar): 8720

h-Index (Google Scholar): 56

i10-Index (Google Scholar): 146

3. Institute of stay and academic interaction outside India

- University of Waterloo, Waterloo, **Canada**-Commonwealth scholar (Sep. 2010-Feb. 2011)
- Nanyang Technological University, **Singapore** Postdoctoral Fellow (Aug 2011-Sept 2012)
- McGill University, **Canada-** Postdoctoral Fellow (Sept 2012-Dec 2012)
- University of Nebraska, Lincoln, **USA-** WARI Fellow (May 2016-Aug 2016)
- Aristotle University of Thessaloniki, **Greece** –Erasmus Fellow (June-July 2015, 2017)
- McGill University, **Canada-** Visiting Scholar (May 2019-June 2019)

Education:

- B Tech, Civil Engineering, National Institute of Technology, Warangal, 2004
- M Tech, Civil Engineering, Indian Institute of Technology, Roorkee, 2006
- PhD, Civil Engineering, Indian Institute of Technology, Roorkee and jointly with University of Waterloo, Waterloo, Canada, 2011
- Postdoctoral Fellow, Civil Engineering, Nanyang Technological University, Singapore, 2012
- Postdoctoral Fellow, Bioresource Engineering, McGill University, Canada, 2012

4. Professional Experience: 15 years of Teaching, Research and Consultancy

Sl. No.	Organization	Designation	Period
1.	Indian Institute of Technology, Indore	Professor	17 Feb 2022 onwards
		BIS Chair Professor	19 July 2023 onwards
		Associate Professor	29 Nov 2017-16Feb 2022
2.	Indian Institute of Technology, Guwahati	Assistant Professor	10 Dec 2012-28 Nov 2017
3.	McGill University, Canada	Postdoctoal Fellow	Sept 2012-Dec 2012
4.	Nanyang Technological University, Singapore	Research Fellow-postdoctoral	Aug 2011-Sept 2012
5.	University of Waterloo, Waterloo, Canada	Research Assistant	Sept 2010-Feb 2011
6.	CNIA Pvt. Ltd., Bangalore	Manager	April 2008-July 2009
7.	Jones Lang LaSalle Pvt. Ltd., Chennai and New Delhi	Assistant Manager	June 2006-April 2008

Administration:

- Dean, Infrastructure Development| 15 June 2020- 19 June 2025
- Associate Dean, Infrastructure Development| 14 March 2019-14 June 2020

5. Academic Awards/ Recognitions/Achievements:

- Guest Professor, University of Notre Dame, USA.
- Editor- PLOS-Climate, Scientific Reports
- Ranked in **Top 2%** scientist in list prepared by Stanford University and among the top 500 of the world's more than 42000 researchers in the sub-area "environmental engineering (Meteorology and Atmospheric Sciences)- 2020-25, **Six times in succession**.
- Chair Professor- BIS Standardization (First Chair Professor of the Institute)

- PI and Coordinator, Centre for Narmada River basin Management studies, Govt of India
- PI and Coordinator, DST-Centre for Policy- Water and Climate, Govt of India
- Prof S N Gupta Memorial Lecture Award, Indian Society of Hydraulics, 2021
- Associate Editor: ASCE- J. of Hazardous, Toxic and Radioactive Waste, Int. J. of Cloud Applications and Computing, Int. J. of Software Science and Computational Intelligence
- Best Researcher of the Year Award 2020 by Institution of Engineers (India)
- RSRI Young Researcher Award 2020
- Associate Editor- ASCE- Journal of Hazardous, Toxic and Radioactive Waste
- Recipient of American Society of Civil Engineers (ASCE's) Environmental and Water Resources Institute (EWRI) Visiting International Fellowship (VIF) -2019
- Prof R J Garde Research Award, Indian Society of Hydraulics, 2019
- Recipient of American Society of Civil Engineers (ASCE)-Best Theoretical-Oriented Paper Award 2018
- NVIDIA GPU Research Grant 2018
- BRNS Young Scientist Research Grant, 2017

6. Sponsored Research Projects: 18 Completed, 03 Ongoing
Consultancy Project: 20 Completed

7. PhD Research Students supervised: 13 Completed, 09 Ongoing

8. Training Program /Workshop/Conferences Organized: 45

9. List of Books

1) Roy Srija and **Goyal Manish Kumar,** *Environmental Monitoring with Integrated Earth Observation Data and Machine Learning*, **Springer Nature**, May 2026, ISBN: **978-3-032-19934-8**. https://doi.org/10.1007/978-3-032-19935-5
2) **Goyal Manish Kumar,** Jain Vijay and Yadav Utkarsh**,** Monitoring India's Ramsar Wetlands: Water Quality and Ecosystem Health via Remote Sensing and AI, Springer, July 2025, ISBN: 978-3-031-96818-1. https://doi.org/10.1007/978-3-031-96818-1
3) **Goyal Manish Kumar**, Shivukumar Rakkasagi, Key Biodiversity Areas: Technology–policy Interface for Combating Forest Loss and Fires, Springer, March-2025, ISBN: 9783031848193 https://doi.org/10.1007/978-3-031-84820-9
4) **Goyal Manish Kumar**, Sachidanand Kumar, Akhilesh Gupta, AI Innovation for Water Policy and Sustainability, Oct-24, Springer Cham, https://doi.org/10.1007/978-3-031-72014-7

5) **Goyal Manish Kumar**, Kuldeep Singh Rautela, Aerosol Atmospheric Rivers Availability, Spatiotemporal Characterisation, Predictability, and Impacts, Jul-24, Springer Cham, https://doi.org/10.1007/978-3-031-66758-9
6) **Goyal Manish Kumar**, Shivam Singh, Understanding Atmospheric Rivers Using Machine Learning, Jun-24, Springer Cham, http://dx.doi.org/10.1007/978-3-031-63478-9
7) **Goyal Manish Kumar** and A. Gupta, 2023 Ecosystem Restoration: Towards Sustainability and Resilient Development, Springer. https://doi.org/10.1007/978-981-99-3687-8
8) **Goyal Manish Kumar**, A.K. Gupta, and A. Gupta., 2022. Hydro-Meteorological Extremes and Disasters, Springer. https://doi.org/10.1007/978-981-19-0725-8
9) Rao Y Surampalli, T.C. Zhang, **Goyal Manish Kumar**, S. K. Brar, R.D. Tyagi., 2020. Sustainability: Fundamentals and applications. John Wiley & Sons. https://doi.org/10.1002/9781119434016
10) **Goyal Manish Kumar,** A.K. Gupta, and A. Gupta., 2020. Integrated Risk of Pandemic: Covid-19 Impacts, Resilience, and Recommendations. Disaster Resilience and Green Growth. Singapore: Springer Singapore. https://doi.org/10.1007/978-981-15-7679-9
11) **Goyal Manish Kumar** 2017. Engineering Hydrology. PHI Learning Pvt. Ltd., New Delhi, India. ISBN: 978-81-203-5243-8
12) **Goyal Manish Kumar** 2015. Fluid Mechanics and Hydraulic Machines. PHI Learning Pvt. Ltd., New Delhi, India. ISBN: 978-81-203-5117-2

10. Technology Transferred/Patents, If any. (State whether patents were commercialized).

Granted Patent

1. Vijay Jain, Utkarsh Yadav, **Manish Kumar Goyal**, "System and a Method for Monitoring Water Quality of a Wetland". Patent Granted: 576312. **Year:** 2026.

Status: GRANTED, **Commercialization:** Under process with MP State Wetland Authority

2. Vijay Jain, Utkarsh Yadav, and **Manish Kumar Goyal**. "*Method and System for Risk Assessment and Management for a Wetland"*. Patent Granted: 589337. Year: 2026.

Filed Patents

1. **Manish Kumar Goyal**, Jhilam Sinha, and Jew Das.
   A Method for Measuring Hydrological Resilience of a Region in Response to Warming Shifts.
   India Patent Application No.: 201921037336, **Year:** 2021.
   Status: Filed | **Commercialization:** Not reported
2. Shivam Singh and **Manish Kumar Goyal**.
   Method and System for Determining Spatiotemporal Characteristics and Predicting the State of Atmospheric Rivers (AR).
   India Patent Application No.: 202421010646, **Year:** 2024.
   Status: Filed | **Commercialization:** Not reported
4. Vijay Jain, Sarthak Agrawal, and **Manish Kumar Goyal**.
   System and Method for Assessing Flood Risk.
   India Patent Application No.: Not available, **Year:** 2025.
   Status: Filed | **Commercialization:** Not reported
5. Kuldeep Singh Rautela and **Manish Kumar Goyal.**
   System and Method for Forecasting Air Quality.

India Patent Application No.: 202521120675, **Year:** 2025.
Status: Filed | **Commercialization:** Under process with Madhya Pradesh Pollution Control Board

6. **Manish Kumar Goyal,** Srija Roy, and Shivam Singh.
Riparian Zone Extractor, Mapper, and Condition Assessment Tool.
India Patent Application No.: 202521130273, **Year:** 2025.
Status: Filed | **Commercialization:** Not reported
7. Pradip Kailas Gopal, Vijay Jain, and **Manish Kumar Goyal**.
A Method and System for Conducting Water Audit in Water Utilities and Industries Across India.
India Patent Application No.: Not available, **Year:** 2026.
Status: Filed | **Commercialization:** Not reported
8. Adithya G, **Kuldeep Singh Rautela**, Mayank Upadhyay, and Manish Kumar Goyal.
Development and Validation of a Low-Cost Air Quality Sensor.
Status: Filed | **Commercialization:** Not reported
9. Srija Roy and **Manish Kumar Goyal.**
Automated Urban Water Body Analyzer.
Status: Filed | **Commercialization:** Under process with MP Water Resource Department
10. Srija Roy and **Manish Kumar Goyal.**
Automated Extraction of Training Samples and Land Use/Land Cover Classification.
Status: Filed | **Commercialization:** Not reported
11. Vijay Jain, Shiv Patel, and **Manish Kumar Goyal.**
Water Resources Assessment for Regional Flood and Droughts.
Status: Filed | **Commercialization:** Not reported
12. Srija Roy, Deepak Mishra, and **Manish Kumar Goyal.**
System and Method for Urban Livability and Investability Assessment Using Multi-Source Earth Observation Data and Predictive Geospatial Modeling.
Status: Filed | **Commercialization:** Not reported

Technology Transfer in advance stage

1. Patent Granted: 576312- Wetland ecosystems especially Ramsar Wetlands are critical for ecological sustainability, yet the lack of real-time water quality monitoring limits effective management and policy intervention. In response, a comprehensive system for wetland water quality monitoring has been developed, enabling near real-time data-driven assessment of wetland health with help of advance remote sensing techniques. The technology is in advance stage of deployment with MP State Wetland Authority.
2. Patent Granted: 589337 – Method and System for Risk Assessment and Management for a Wetland. This patented innovation provides a comprehensive framework for assessing and managing wetland risks by integrating environmental indicators, hazard assessment, and decision-support methodologies. The system facilitates proactive planning, enhances ecosystem resilience, and supports sustainable wetland governance. The technology is also being advanced for implementation in collaboration with the Madhya Pradesh State Wetland Authority.
3. India Patent Application No.: 202521120675- Urban areas of Madhya Pradesh face significant limitations in real-time air quality monitoring, and reliable forecasting systems are currently lacking. To address this gap, a comprehensive air quality monitoring and forecasting tool (AeroVision) has been developed for urban centers in accordance with NAAQS guidelines. The system provides forecasts up to six days in advance by integrating ensemble artificial intelligence techniques with ground-based observations, along with health advisory warnings. The technology has reached an advanced stage of development and is currently under process for technology transfer to the Madhya Pradesh Pollution Control Board (MPPCB), Government of Madhya Pradesh.

4. Patent Filed: Urban water bodies are increasingly vulnerable to pollution due to rapid urbanization and inadequate monitoring systems. To address this challenge, an Automated Urban Water Body Analyzer has been developed, providing a data-driven and scalable solution for real-time assessment of water bodies and water quality with hotspot identification. The technology has been filed and is currently under transfer stage with the Madhya Pradesh Water Resources Department for operational deployment.

11. Complete list of research publications:

1) Kuldeep Singh Rautela, **Manish Kumar Goyal**, and Mohan M., *A review of extreme air pollution measurement and modeling techniques with applications*, Renewable and Sustainable Energy Reviews, 240, 117178, 2026. IF: 18.0
2) Amitoj Raj, Kuldeep Singh Rautela, and Manish Kumar Goyal, *Climate Change, Wetland Resilience, and the Sustainable Development Goals in Central India: Implications for Water Security and Adaptation Policy*, Sustainable Development, 2026. IF: 9.3
3) Shivukumar Rakkasagi, Kushal Jagdish Patil, **Manish Kumar Goyal**, and S. Maman, *Flood Risk in India's Arid Ramsar Wetlands: Integrating Extreme Rainfall, Fuzzy Risk Assessment, and CMIP6 Projections*, Physics and Chemistry of the Earth, Parts A/B/C, 104522, 2026. IF: 4.6
4) Kuldeep Singh Rautela, Mayank Upadhyay, **Manish Kumar Goyal**, and S. Taigor, *Artificial Intelligence–Driven Air Quality Mapping and Forecasting for Extreme PM2.5 Pollution Events*, Physics and Chemistry of the Earth, Parts A/B/C, 104459, 2026. IF: 4.6
5) Deepak Mishra, Nakka Naveen Kumar, Shekhar Singh, and **Manish Kumar Goyal**, The future of global Ramsar wetlands under intensifying precipitation extremes: Arid regions as emerging hotspots, Environmental Impact Assessment Review, 118, 108275, 2026. https://doi.org/10.1016/j.eiar.2025.108275, IF: 11.2
6) Shashank Agrawal, Shivukumar Rakkasagi, and **Manish Kumar Goyal**, High-resolution landfill characterization using SAR remote sensing and cloud-based processing, Scientific Reports, 2025. https://doi.org/10.1038/s41598-025-32908-9 IF: 3.9
7) Kuldeep Singh Rautela, **Manish Kumar Goyal**, and Ajay Singh Nagpure, Unequal spatio-temporal distribution of population-weighted pollution extremes through deep learning, npj Climate and Atmospheric Science 8(1), 340, 2025, https://doi.org/10.1038/s41612-025-01183-w IF: 8.4
8) Shivukumar Rakkasagi, **Manish Kumar Goyal**, and Srinidhi Jha, A techno- societal framework for quantifying the risk of heavy rainfall events over Ramsar wetlands. Journal of Geophysical Research: Oceans, 130(9), e2025JC022563, 2025, https://doi.org/10.1029/2025JC022563, IF: 3.4
9) Shivukumar Rakkasagi, Stuart G. Edris, and **Manish Kumar Goyal**, A socio-technical framework for assessing flash drought risks to key biodiversity areas: A scientists' warning, Technology in Society, 82, 102893, 2025, https://doi.org/10.1016/j.techsoc.2025.102893, IF: 12.5
10) Shekhar Singh, Vijay Jain, and **Manish Kumar Goyal**, Evaluating climate shifts and drought regions in the central Indian river basins, *Scientific Reports* 15(1), 29701, 2025, https://doi.org/10.1038/s41598-025-15231-1, IF: 3.9

11) Shivukumar Rakkasagi, and **Manish Kumar Goyal**. Protecting wetlands for future generations: A comprehensive approach to the water-climate-society nexus in South Asia, Environmental Impact Assessment Review, 115, 107988, 2025, https://doi.org/10.1016/j.eiar.2025.107988, IF: 11.2
12) Vijay Jain, Shekhar Singh, and **Manish Kumar Goyal**, Corrigendum: Urban precipitation dynamics risk assessment across Indian smart cities (2025 Environ. Res. Lett. 20 074059), Environmental Research Letters, 20(8), 089502, 2025, https://doi.org/10.1088/1748-9326/ade3f0, IF: 5.6
13) C.S.P. Ojha, V. P. Singh, **Manish Kumar Goyal**, Ashish Sharma, and Deepak Kumar Tiwari. "Futuristic and Sustainable Aspects in Engineering and Technology (FSAET-2023)." *Journal of Water and Climate Change* 16(8), iii-v, 2025. [IF 3.0].
14) Kuldeep Singh Rautela, **Manish Kumar Goyal**, and Rao Y. Surampalli, AI and Machine Learning for Optimizing Waste Management and Reducing Air Pollution, Journal of Hazardous, Toxic, and Radioactive Waste, 29(3), 04025014, 2025, https://doi.org/10.1061/JHTRBP.HZENG-1483, IF: 2.9.
15) Kuldeep Singh Rautela, Nidhi Kanwar, Jagdish Chandra Kuniyal, Mohd Sharjeel Sofi, **Manish Kumar Goyal**, Ajay Singh Bhoj, Sami Ullah Bhat et al. Assessing Hydro-climatological Variability and Land Use Characteristics of the Headwater Basins of the Indian Himalayan Region, Earth Systems and Environment, 9, 2211-2233, 2025, https://doi.org/10.1007/s41748-025-00662-7, IF: 4.7
16) Shivukumar Rakkasagi, and **Manish Kumar Goyal**.Are rapid onset drying events escalating forest fires across India's ecoregions?. Environmental Research Letters, 20(6), 064004, 2025 https://doi.org/10.1088/1748-9326/add1f5, IF: 5.6.
17) Vijay Jain, Sachidanand Kumar, **Manish Kumar Goyal**, Relationship between daily precipitation extremes and temperature in changing climate across smart cities of Central India, Journal of Environmental Management, 380, 1125036, 2025, https://doi.org/10.1016/j.jenvman.2025.125036, IF: 8.4
18) Shekhar Singh**, Manish Kumar Goyal**, A review of India's water policy and implementation toward a sustainable future, Water and Climate Change, 16 (2): 493–510, 2025, https://doi.org/10.2166/wcc.2025.560, IF: 2.8
19) Kuldeep Singh Rautela, **Manish Kumar Goyal**, Modelling Health Implications of Extreme PM2.5 Concentrations in Indian Sub-Continent: Comprehensive Review with Longitudinal Trends and Deep Learning Predictions, Technology in Society, Volume 81, June 2025, 102843, https://doi.org/10.1016/j.techsoc.2025.102843, IF: 12.5
20) Saket Dubey, Ashim Sattar, Vivek Gupta, **Manish Kumar Goyal**, Umesh K. Haritashya, Jeffrey S. Kargel, Transboundary hazard and downstream impact of glacial lakes in Hindu-Kush Karakoram Himalayas, Science of The Total Environment., 914(1), 2025, 169758, https://doi.org/10.1016/j.scitotenv.2023.169758, IF: 8.2
21) Raghavendra Nargund, Kuldeep Singh Rautela, **Manish Kumar Goyal**, Nishant K Sinha, Monoranjan Mohanty, Virender Singh Bhatia, Assessing soybean yield in Madhya Pradesh by using a multi-model approach, Field Crops Research, 322, 109716, 2025, https://doi.org/10.1016/j.fcr.2024.109716, IF: 6.4
22) Kuldeep Singh Rautela, **Manish Kumar Goyal**, Spatio-temporal analysis of extreme air pollution and risk assessment, Journal of Environmental Management, Volume 373, 2025, 123807, https://doi.org/10.1016/j.jenvman.2024.123807, IF: 8.4.

23) Rajneesh Kumar, **Manish Kumar Goyal**, Role of AI&ML in Modernizing Water and Wastewater Treatment Processes, Water, Air, & Soil Pollution, Volume 236, December 2024, 1-22, https://doi.org/10.1007/s11270-024-07618-z, IF: 3.8.
24) **Manish Kumar Goyal**, Shivukumar Rakkasagi, Rao Y. Surampalli, Tian C. Zhang, Saikumar Erumalla, Abhijeet Gupta, Saket Dubey, Chalida U-tapao , Enhancing Sustainable Development through Spatiotemporal Analysis of Ramsar Wetland Sites in South Asia, Technology in Society, Volume 79, December 2024, 102723, https://doi.org/10.1016/j.techsoc.2024.102723, IF: 10.1
25) Rajneesh Kumar, Avinash Mishra, **Manish Kumar Goyal**, Wastewater management policies: a perspective and roadmap for India., Current Science, Volume 7, October 2024, 795, http://dx.doi.org/10.18520/cs/v127/i7/795-807, IF: 1.
26) Nikhil Kumar, **Manish Kumar Goyal**, Projected changes in monsoonal compound dry-hot extremes in India, Atmospheric Research, Volume 310, November 2024, 107605, https://doi.org/10.1016/j.atmosres.2024.107605 , IF: 4.5
27) Rajneesh Kumar, **Manish Kumar Goyal**, Rao Y Surampalli, Tian C Zhang, River pollution in India: exploring regulatory and remedial paths, Clean Technologies and Environmental Policy, Volume 26, September 2024, pages 2777–2799, https://doi.org/10.1007/s10098-024-02763-9 , IF: 4.2
28) Kuldeep Singh Rautela, **Manish Kumar Goyal**, Transforming air pollution management in India with AI and machine learning technologies, Scientific Reports, September 2024, 20412, https://doi.org/10.1038/s41598-024-71269-7 , IF: 3.8
29) Rajneesh Kumar, Avinash Mishra, **Manish Kumar Goyal**, Water neutrality: Concept, challenges, policies, and recommendations, Groundwater for Sustainable Development, Volume 26, August 2024, 101306, https://doi.org/10.1016/j.gsd.2024.101306 , IF: 4.9
30) Kuldeep Singh Rautela, Shivam Singh, **Manish Kumar Goyal**, Aerosol atmospheric rivers: patterns, impacts, and societal insights, Environmental Science and Pollution Research, August 2024, https://doi.org/10.1007/s11356-024-34625-8 , IF: 5.8.
31) Kuldeep Singh Rautela, Shivam Singh, **Manish Kumar Goyal**, Resilience to Air Pollution: A Novel Approach for Detecting and Predicting Aerosol Atmospheric Rivers within Earth System Boundaries, Earth Systems and Environment, July 2024, https://doi.org/10.1007/s41748-024-00421-0 , IF: 5.3
32) Kuldeep Singh Rautela, Jagdish Chandra Kuniyal, **Manish Kumar Goyal**, Nidhi Kanwar, Ajay Singh Bhoj, Assessment and modelling of hydro-sedimentological flows of the eastern river Dhauliganga, north-western Himalaya, India, Natural Hazards, Volume 120, April 2024, pages 5385–5409, https://doi.org/10.1007/s11069-024-06413-7 , IF: 3.3
33) Shivukumar Rakkasagi, **Manish Kumar Goyal**, Srinidhi Jha, Evaluating the future risk of coastal Ramsar wetlands in India to extreme rainfalls using fuzzy logic, Journal of Hydrology, Volume 632, March 2024, 130869, https://doi.org/10.1016/j.jhydrol.2024.130869 , IF: 5.9
34) Shivam Singh, **Manish Kumar Goyal**, Erumalla Saikumar, Assessing Climate Vulnerability of Ramsar Wetlands through CMIP6 Projections, Water Resources Management, Volume 38, March 2024, pages 1381–1395, https://doi.org/10.1007/s11269-023-03726-3 , IF: 3.9
35) Saket Dubey, Ashim Sattar, Vivek Gupta, **Manish Kumar Goyal**, Umesh K. Haritashya, Jeffrey S. Kargel, Transboundary hazard and downstream impact of glacial lakes in

Hindu-Kush Karakoram Himalayas, Science of The Total Environment, Volume 914, 1 March 2024, 169758, https://doi.org/10.1016/j.scitotenv.2023.169758 , IF: 8.2

36) Kuldeep Singh Rautela, Shivam Singh, **Manish Kumar Goyal**, Characterizing the spatiotemporal distribution, detection, and prediction of aerosol atmospheric rivers on a global scale, Journal of Environmental Management, Volume 351, February 2024, 119675, https://doi.org/10.1016/j.jenvman.2023.119675 , IF: 8
37) Shivam Singh, Aditya Yadav, **Manish Kumar Goyal**, Univariate and bivariate spatiotemporal characteristics of heat waves and relative influence of large-scale climate oscillations over India, Journal of Hydrology, Volume 628, January 2024, 130596, https://doi.org/10.1016/j.jhydrol.2023.130596 , IF: 5.9
38) Raghu Nath Prajapati; Nurazim Ibrahim; **Manish Kumar Goyal**; Bhesh Raj Thapa; Koshish Raj Maharjan, Ground water availability assessment for a data-scarce river basin in Nepal using SWAT hydrological model, Water Supply, December 2023, ws202332, https://doi.org/10.2166/ws.2023.332 , IF: 1.9
39) Samarth Meghani, Shivam Singh, Nagendra Kumar **Manish Kumar Goyal**, Predicting the spatiotemporal characteristics of atmospheric rivers: A novel data-driven approach, Global and Planetary Change, Volume 231, December 2023, 104295, https://doi.org/10.1016/j.gloplacha.2023.104295 , IF: 4
40) Deepti Shakya, Vishal Deshpande, **Manish Kumar Goyal**, Mayank Agarwal, PM2.5 air pollution prediction through deep learning using meteorological, vehicular, and emission data: A case study of New Delhi, India, Journal of Cleaner Production, Volume 427, 15 November 2023, 139278, https://doi.org/10.1016/j.jclepro.2023.139278 , IF: 9.7
41) **Manish Kumar Goyal**, Shivukumar Rakkasagi, Soumya Shaga, Tian C. Zhang, Rao Y. Surampalli & Saket Dubey, Spatiotemporal-based automated inundation mapping of Ramsar wetlands using Google Earth Engine, Scientific Reports, October 2023, 17324, https://doi.org/10.1038/s41598-023-43910-4 , IF:3.8.
42) Saket Dubey, Ashim Sattar, **Manish Kumar Goyal**, Simon Allen, Holger Frey, Umesh K. Haritashya, Christian Huggel, Mass Movement Hazard and Exposure in the Himalaya, Earth's Future, September 2023, https://doi.org/10.1029/2022EF003253 , IF: 7.3
43) **Manish Kumar Goyal**, Shivam Singh & Vijay Jain, Heat waves characteristics intensification across Indian smart cities, Scientific Reports, September 2023, https://doi.org/10.1038/s41598-023-41968-8 , IF: 3.8.
44) Manish Kumar Goyal, Anil Kumar Gupta, Jew Das, Vijay Jain, Shivukumar Rakkasagi, Heatwave magnitude impact over Indian cities: CMIP 6 projections, Theoretical and Applied Climatology, Volume 154, August 2023, 959-971, https://doi.org/10.1007/s00704-023-04599-7 , IF: 2.8
45) Shivukumar Rakkasagi; Vikas Poonia; **Manish Kumar Goyal**, Flash drought as a new climate threat: drought indices, insights from India's study, and implications for future research, Water and Climate Change, August 2023, 14 (9): 3368–3384., https://doi.org/10.2166/wcc.2023.347 , IF: 2.8.
46) Nikhil Kumar, Piyush Patel, Shivam Singh, **Manish Kumar Goyal**, Understanding non-stationarity of hydroclimatic extremes and resilience in Peninsular catchments, India, Scientific Reports, August 2023, https://doi.org/10.1038/s41598-023-38771-w , IF: 3.8
47) **Manish Kumar Goyal**, Vikas Poonia, Vijay Jain, Three Decadal Urban Drought Variability Risk Assessment for Indian Smart Cities, Journal of Hydrology, Volume 625, Part A, October 2023, 130056, https://doi.org/10.1016/j.jhydrol.2023.130056 , IF: 5.9.

48) Shivam Singh, **Manish Kumar Goyal**, Enhancing climate resilience in businesses: The role of artificial intelligence, Journal of Cleaner Production, Volume 418, 15 September 2023, 138228, https://doi.org/10.1016/j.jclepro.2023.138228 , IF: 9.7
49) Moirangthem Johnson Singh, Sourabh Choudhary, Wen-Bo Chen, Pei-Chen Wu, **Manish Kumar Goyal**, Abhishek Rajput, Lalit Borana, Applications of Fibre Bragg Grating Sensors for monitoring geotechnical structures: A Comprehensive Review, Measurement, Volume 218, 15 August 2023, 113171, https://doi.org/10.1016/j.measurement.2023.113171 , IF: 5.2.
50) Shivam Singh, **Manish Kumar Goyal**, An innovative approach to predict atmospheric rivers: Exploring convolutional autoencoder, Atmospheric Research, Volume 289, July 2023, 106754, https://doi.org/10.1016/j.atmosres.2023.106754 , IF: 4.5
51) Shivam Singh, Nikhil Kumar, **Manish Kumar Goyal**, Srinidhi Jha, Relative influence of ENSO, IOD, and AMO over spatiotemporal variability of hydroclimatic extremes in Narmada basin, India, AQUA - Water Infrastructure Ecosystems and Society, March 2023, 72 (4): 520–539, https://doi.org/10.2166/aqua.2023.219 , IF: 2.1
52) Shivam Singh, **Manish Kumar Goyal**, Srinidhi Jha, Role of large-scale climate oscillations in precipitation extremes associated with atmospheric rivers: nonstationary framework, Hydrological Sciences Journal, January 2023, 395-411, https://doi.org/10.1080/02626667.2022.2159412 , IF: 2.8
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Contributions to the System in Leadership Roles

Prof. Manish Kumar Goyal | IIT Indore

Administrative Leadership

- Former Dean (2020–2025), Indian Institute of Technology, Indore — responsible for Infrastructure Development of worth ~1400 crore including development and restoration of 08 water bodies of 5.4 million litres capacity.
- BIS Chair Professor (First Chair Professor of IIT Indore) — established 07 student chapters and awareness program for more than 5000 students and faculty in Central India

National Research Coordination

- Principal Investigator (PI) and Coordinator, Centre for Narmada River Basin Management Studies, Government of India
- PI and Coordinator, DST-Centre for Policy Research – Water and Climate, Government of India steering policy-oriented research linking water resources and climate change at a national level.

Editorial Leadership

- Editor, PLOS-Climate contributing to shaping the direction of climate research publications globally.
- Editor, Scientific Reports (Nature Portfolio) reviewing and guiding multidisciplinary research.
- Associate Editor, ASCE – Journal of Hazardous, Toxic and Radioactive Waste.
- Associate Editor, International Journal of Cloud Applications and Computing.
- Associate Editor, International Journal of Software Science and Computational Intelligence.

Sponsored Research & Consultancy

- Successfully completed 18 sponsored research projects and 3 ongoing, demonstrating sustained contribution to national R&D priorities.
- 20 consultancy projects completed by providing technical expertise to industry and government.

Human Resource Development

- Supervised 13 PhD scholars to completion and 9 ongoing significant contribution to building India's research capacity in water, climate, and environmental engineering.
- Organized 45 training programs, workshops, and conferences fostering knowledge transfer and capacity building nationally and internationally, trained more than 4500 students

Technology & Innovation

- Holds 17 patents (2 granted, 15 filed) in water quality monitoring, hydrological resilience, atmospheric river prediction, air quality forecasting and sensor development, and flood risk assessment bridging research with applied solutions.

International Collaboration & Representation

- Led collaborative research with institutions across Canada, USA, Singapore, Greece, and other countries, representing India in global scientific activities.
- JSPS Fellow, Commonwealth Fellow, Erasmus Fellow, Indo-US WARI Fellow recognized for international scientific contributions.

- **Societal Impact of Research Work**

- Prof. Goyal's research on water resources, climate change, and hazard modelling informs national water and climate policy in India. His AI-driven frameworks for drought, flood, and air quality prediction help policymakers and communities prepare for climate extremes. As PI of Government of India centres on Narmada basin management and policy research, he advances governance and sustainability. His studies on wetlands, glacial lake outburst floods, and urban extremes provide actionable risk assessments. The AeroVision application delivers real-time pollutant concentrations, AQI, and six-day forecasts, enabling early warnings, reducing exposure risks, supporting public health decisions, and strengthening government pollution control planning.